# Reproducible Data Analysis

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# 1 Busiest Airports Analysis

## 1.1 Overview

This section examines passenger traffic trends across six major international airports from 2020 to 2024. The years covered span the COVID-19 pandemic's impact on air travel through the

subsequent recovery period. The six airports selected — Atlanta (ATL), Frankfurt (FRA), Beijing Daxing (PKX), Dubai (DXB), Los Angeles (LAX), and London Heathrow (LHR) — represent key hubs across North America, Europe, the Middle East, and Asia.

## 1.2 Passenger Traffic Table

Table 1: Annual Passenger Traffic for Six Major Airports (2020-2024)

| Airport | 2020 | 2021 | 2022 | 2023 | 2024 |
|---|---|---|---|---|---|
| Atlanta | 42,918,685 | 75,704,760 | 93,699,630 | 104,653,451 | 108,067,766 |
| Frankfurt | 18,768,601 | 24,812,849 | 48,918,482 | 59,355,389 | 61,564,957 |
| Beijing Daxing | N/A | 25,051,012 | N/A | 39,400,000 | 49,441,029 |
| Dubai | 25,836,771 | 29,110,609 | 66,069,981 | 86,994,365 | 92,331,506 |
| Los Angeles | 28,779,527 | 48,007,284 | 65,924,298 | 75,050,875 | 76,588,028 |
| London Heathrow | 22,111,469 | N/A | 61,614,508 | 79,183,364 | 83,884,572 |

The table reveals the dramatic collapse of passenger traffic in 2020 due to the COVID-19 pandemic, followed by a strong and consistent recovery through 2024. Atlanta consistently led all airports in total passengers, reaching over 108 million in 2024. Beijing Daxing (PKX) did not appear in the top 50 in 2020 and 2022, reflecting China's strict border restrictions during those years. London Heathrow similarly fell outside the top 50 in 2021 due to the UK's travel restrictions.

### 1.3 Passenger Traffic Over Time

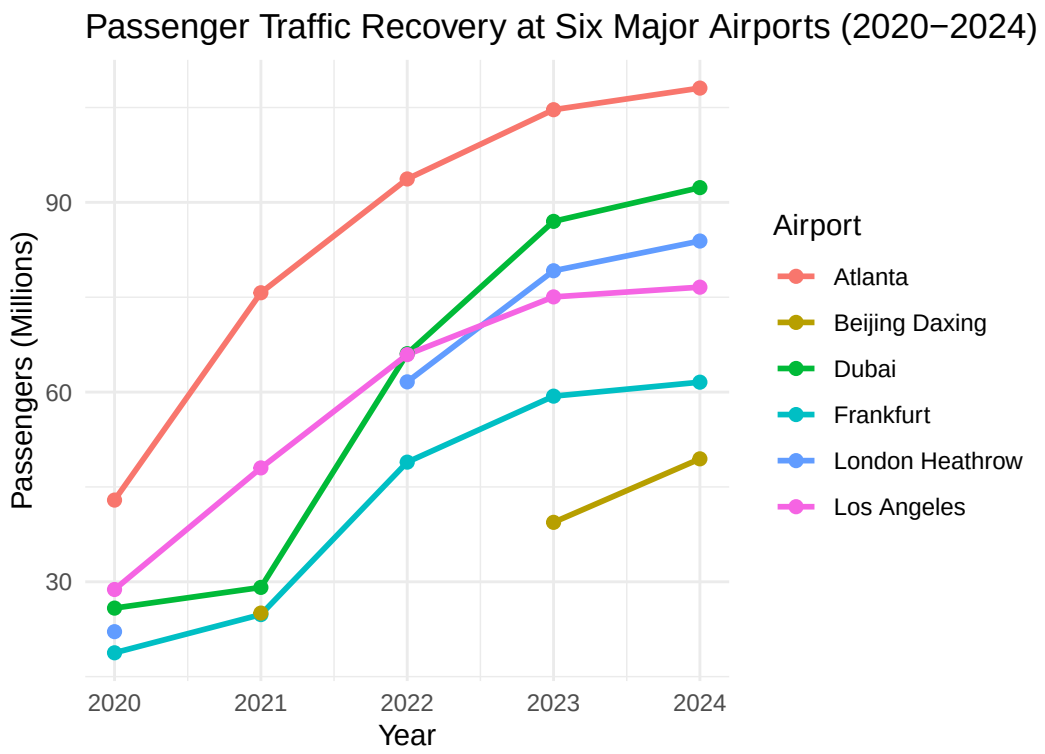


Figure 1: Passenger traffic (in millions) for six major airports, 2020-2024.

The line graph makes the recovery trajectory clear for each airport. All six airports experienced sharp declines in 2020, with Dubai and Heathrow among the hardest hit. By 2023 and 2024, most airports had not only recovered but surpassed pre-pandemic levels. Atlanta's consistent dominance is visually apparent, while Beijing Daxing shows the steepest growth curve as it ramped up operations from a near-standstill.

## 2 Monte Carlo Numerical Integration

### 2.1 Method Overview

Monte Carlo numerical integration estimates the value of a definite integral using random sampling. The approach works by generating random points within a rectangular window that bounds the function, then calculating the proportion of points that fall at or below the curve.

Multiplying this proportion by the area of the rectangle yields an estimate of the integral. As the number of sampled points increases, the estimate converges toward the true value.

For this analysis, the function being integrated is the probability density function of the Beta distribution with shape parameters $\alpha = 2$ and $\beta = 2$, integrated over $x \in [0, 1]$ with a rectangular window of $y \in [0, 1.5]$.

## 2.2 Small Multiple Visualization

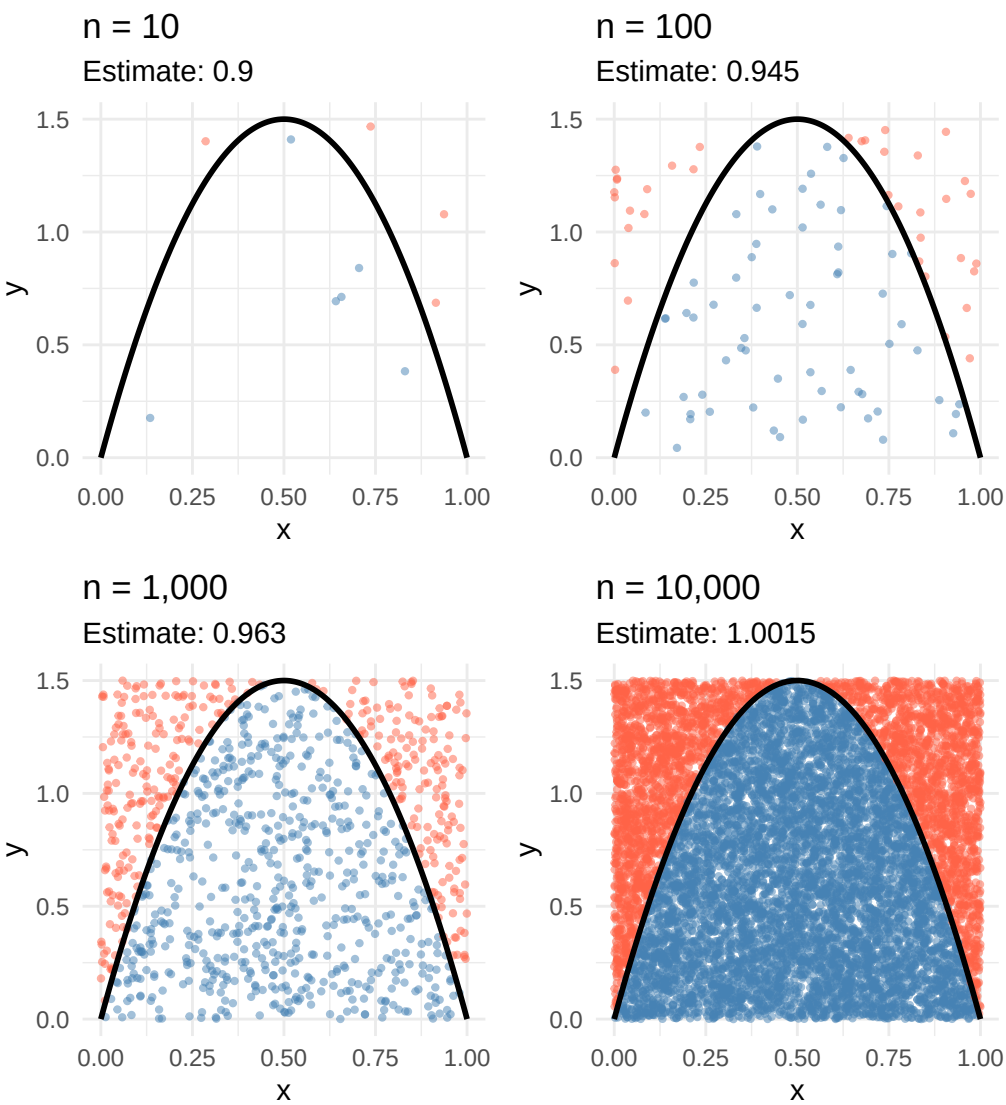


Figure 2: Monte Carlo integration of the Beta(2,2) PDF at four resolutions.

## 2.3 Results and Discussion

The small multiple visualization demonstrates how the Monte Carlo estimate improves with more simulated points. At $n = 10$, the estimate is highly variable and unreliable. By $n = 1{,}000$ and $n = 10{,}000$, the estimate stabilizes and approaches the true value of 1.0, which is expected

since the Beta distribution PDF integrates to exactly 1 over $[0, 1]$. This illustrates a core principle of Monte Carlo methods: precision improves with resolution.

# 3 Planning and Prompting GenAI Tools

## 3.1 Data Context and Plan

**Goals:** The calcium dataset is currently in a wide format, which makes it harder to analyze trends properly. I want to restructure it so that each observation is in its own row, making the data tidy and easier to work with. My main focus is to track how calcium levels change over time and see if diet and exercise play a role.

**Needs:**

- calcium.csv dataset
- null treatment group, diet/exercise treatment group
- Variables: Initial, Year 1, Year 2, Year 3
- Packages: tidyverse, ggplot2
- Functions: read_csv(), pivot_longer(), mutate(), bind_rows(), ggplot(), geom_line()

**Steps:**

1. Load calcium.csv using read_csv()
2. Rename columns to identify which group they belong to
3. Separate the data into two group-specific data frames
4. Use pivot_longer() to reshape each group from wide to long format
5. Add a group label column to each reshaped data frame
6. Combine both groups into one tidy data frame using bind_rows()
7. Calculate mean calcium levels per group per year
8. Create a line plot using ggplot2 showing trends over time for each group

## 3.2 Plan-Based Prompt Results

I provided Claude with my plan and the calcium.csv file. Here is the response generated:

```r
library(tidyverse)

# Load the calcium data
calcium <- read_csv("calcium.csv")

# Rename columns to identify groups
colnames(calcium) <- c("null_y0", "null_y1", "null_y2", "null_y3",
```

```r
                        "diet_y0", "diet_y1", "diet_y2", "diet_y3")

# Separate and reshape null treatment group
null_group <- calcium %>%
  select(starts_with("null")) %>%
  drop_na() %>%
  mutate(id = row_number(), group = "Null Treatment") %>%
  pivot_longer(cols = starts_with("null"),
               names_to = "year", values_to = "calcium") %>%
  mutate(year = recode(year, "null_y0" = 0, "null_y1" = 1,
                       "null_y2" = 2, "null_y3" = 3))

# Separate and reshape diet/exercise group
diet_group <- calcium %>%
  select(starts_with("diet")) %>%
  drop_na() %>%
  mutate(id = row_number(), group = "Diet/Exercise") %>%
  pivot_longer(cols = starts_with("diet"),
               names_to = "year", values_to = "calcium") %>%
  mutate(year = recode(year, "diet_y0" = 0, "diet_y1" = 1,
                       "diet_y2" = 2, "diet_y3" = 3))

# Combine groups
calcium_tidy <- bind_rows(null_group, diet_group)

# Summarise mean calcium by group and year
calcium_summary <- calcium_tidy %>%
  group_by(group, year) %>%
  summarise(mean_calcium = mean(calcium, na.rm = TRUE),
            .groups = "drop")

# Plot
ggplot(calcium_summary,
       aes(x = year, y = mean_calcium,
           color = group, group = group)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  labs(
    title = "Mean Ulnar Calcium Levels Over Time by Treatment Group",
    x = "Year",
    y = "Mean Calcium Level",
    color = "Treatment Group"
```

```r
  ) +
  scale_x_continuous(breaks = 0:3,
    labels = c("Initial", "Year 1", "Year 2", "Year 3")) +
  theme_minimal()
```

### 3.3 Generic Prompt Results

The following plots were generated by Anthropic Claude using a generic prompt without a detailed plan. The prompt simply asked Claude to analyze the calcium dataset and create a visualization without any context or structured guidance.

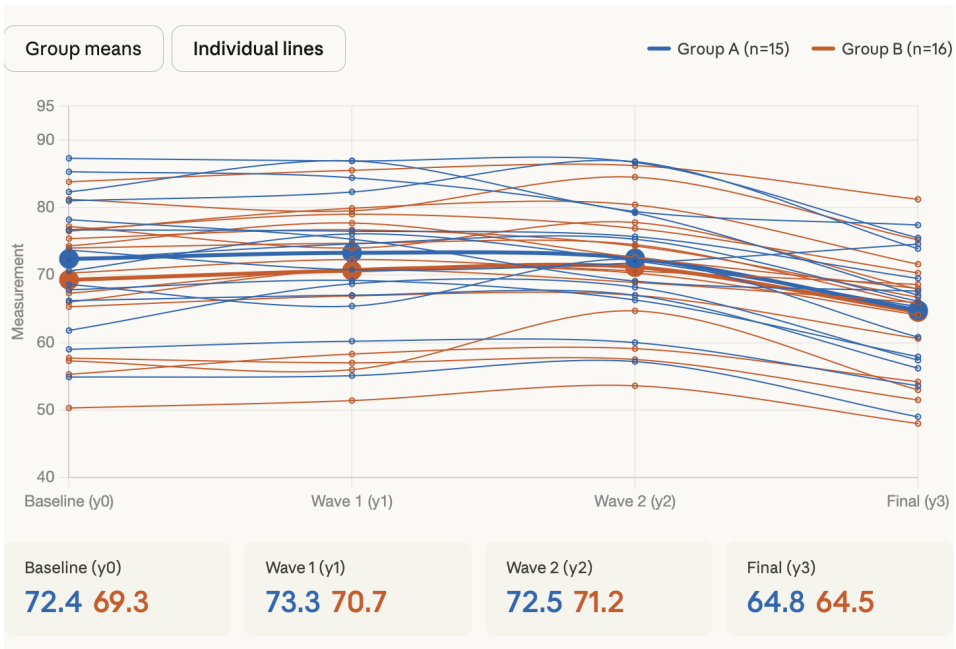


Figure 3: Group means from generic Claude prompt

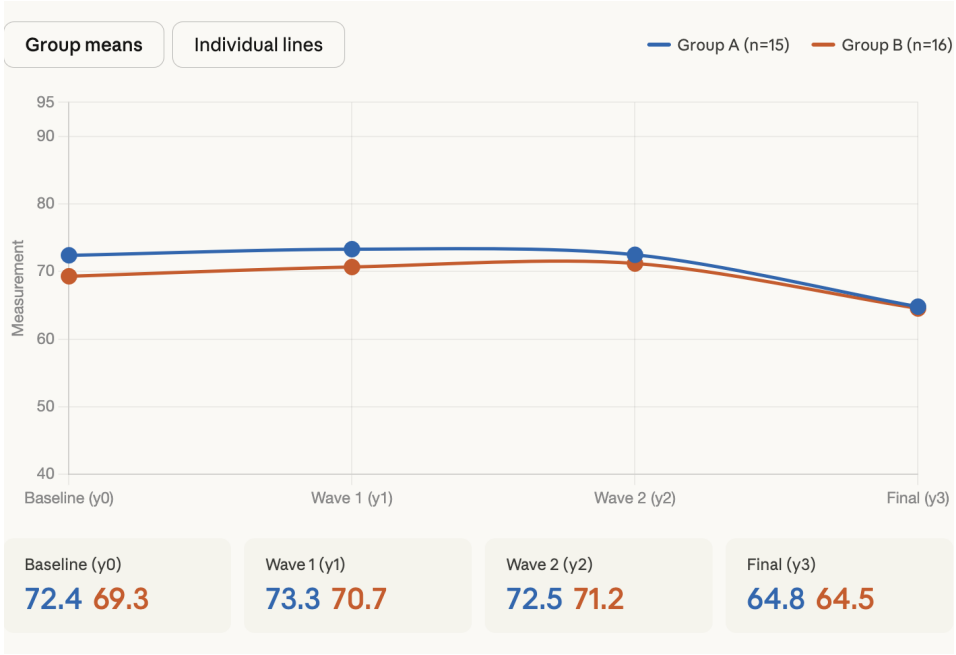


Figure 4: Individual lines from generic Claude prompt

The generic response labeled the groups as Group A and Group B rather than using the meaningful names Null Treatment and Diet/Exercise. Time points were labeled as Wave 1 and Wave 2 rather than Year 1 and Year 2, indicating the tool did not understand the study context without guidance.

### 3.4 Comparison

When comparing the plan-informed response and the generic response, several important differences emerge. The plan-informed response produced code that correctly identified the two groups as Null Treatment and Diet/Exercise, used meaningful variable names, and structured the visualization around the specific research question of tracking calcium levels over time. The generic response, by contrast, labeled the groups simply as Group A and Group B, showing that without context, the tool could not interpret the meaning behind the data structure.

The visualizations themselves also differed significantly. The plan-informed response produced a clean line plot of group means that directly addressed the goal of comparing trends between treatment groups. The generic response produced a more complex plot showing both individual lines and group means, which while visually detailed, was harder to interpret and did not clearly communicate the research question. The generic plot also used labels like Wave 1 and Wave 2 instead of Year 1 and Year 2, suggesting it did not fully understand the study context without guidance.

This exercise reveals an important lesson about working with GenAI tools: the quality of the output is directly tied to the quality of the input. A detailed, context-specific plan guided Claude to produce code and visualizations aligned with the actual research goals. A vague prompt produced technically functional but contextually weak results. This shows that GenAI tools are most powerful when used as collaborators guided by human understanding, not as replacements for it.

# 4 Reflection

One of the most valuable things I have learned in this course is the concept of tidy data. Before this course, I would not have thought twice about a dataset having multiple columns representing the same variable across different time points. Working with the calcium dataset made this concrete — the data had eight columns when it should have had one row per observation. Learning to use pivot_longer() to reshape data has changed how I approach any dataset I encounter, and I now instinctively ask whether each row represents one observation.

I have also grown significantly in my ability to create and interpret data visualizations using ggplot2. Early in the course, I struggled to understand why certain plot types were more appropriate than others. Through the busiest airports analysis, I learned that line graphs are ideal for showing change over time, and that thoughtful labels, captions, and color choices make a visualization genuinely useful rather than just decorative. I also learned the importance of writing narrative text that goes beyond describing what is visible — good analysis explains what the patterns mean and why they matter.

Finally, the Monte Carlo integration exercise taught me that randomness can be a powerful computational tool. I found it surprising that simply generating random points and checking whether they fall below a curve can estimate a definite integral. Watching the estimate converge toward 1.0 as the number of points increased from 10 to 10,000 made the concept of statistical precision feel real and tangible. This course has shown me that data analysis is not just about running code — it is about thinking carefully, planning deliberately, and communicating clearly.

# Appendices

## GenAI Usage

**Tool:** Anthropic Claude (claude.ai, Claude Sonnet 4.6)

**Date:** April 14, 2026

**Prompt 1:** I provided Claude with the calcium.csv file and the following plan: The calcium dataset is currently in a wide format. I want to restructure it so each observation is in its own

row. My goal is to track how calcium levels change over time and whether diet and exercise play a role. I will use pivot_longer() to reshape the data, create a grouping variable, and use ggplot2 to build a line plot showing trends over time for each treatment group. I asked Claude to generate R code based on this plan.

**Response 1:** Claude generated R code that loaded the calcium.csv file, renamed the eight columns to identify null and diet/exercise groups, separated the data into two group-specific data frames, used pivot_longer() to reshape each from wide to long format, combined them with bind_rows(), calculated group means per year, and created a line plot using ggplot2 showing mean ulnar calcium levels over time for each treatment group.

## Code Appendix